OULAD-Based Statistical Analysis

Hypothesis: Positive Learning Experiences Contribute to Sustained Academic Success

Statistics Module — Group Assignment

Complete Technical & Planning Guide

1. OULAD Dataset — Files & Attributes

The Open University Learning Analytics Dataset (OULAD) is a publicly available dataset released by the Open University (UK). It contains data about courses, students, and student interactions with the Virtual Learning Environment (VLE) for 22 courses over multiple years. The dataset is distributed across 7 interrelated CSV files.

1.1 studentInfo.csv

This is the central student demographic and outcome file. Each row represents one student registration in one module presentation.

Attribute	Type	Description
code_module	Categorical	Module identifier (e.g., AAA, BBB). Represents the course.
code_presentation	Categorical	Presentation identifier combining year and semester (e.g., 2013J, 2014B).
id_student	Integer	Unique student identifier.
gender	Binary	Student gender: M or F.
region	Categorical	Geographic region of the student in the UK.
highest_education	Ordinal	Highest level of education prior to registration: No Formal quals → Post Graduate Qualification.
imd_band	Ordinal	Index of Multiple Deprivation — a UK socioeconomic measure in 10% bands.
age_band	Ordinal	Age band: 0–35, 35–55, 55≤.
num_of_prev_attempts	Integer/Discrete	Number of times the student has previously attempted this module.
studied_credits	Integer	Total credits the student is studying in the current year.
disability	Binary	Whether the student has a declared disability: Y or N.
final_result	Ordinal/Categorical	Outcome: Distinction, Pass, Fail, or Withdrawn.

1.2 studentVle.csv

Records every student interaction with VLE resources, logged per day. This is the largest file and is critical for measuring engagement (a proxy for positive learning experiences).

Attribute	Type	Description
code_module	Categorical	Module identifier.
code_presentation	Categorical	Presentation identifier.
id_student	Integer	Student identifier.
id_site	Integer	VLE resource/activity identifier (links to vle.csv).

Attribute	Type	Description
date	Integer	Days relative to module start date (negative = before module start).
sum_click	Integer/Discrete	Number of times the student clicked the resource on that day.

1.3 vle.csv

Describes each VLE resource/activity that students interact with.

Attribute	Type	Description
id_site	Integer	Unique VLE resource identifier.
code_module	Categorical	Module the resource belongs to.
code_presentation	Categorical	Presentation the resource belongs to.
activity_type	Categorical	Type of VLE activity: forumng, oucontent, quiz, resource, url, page, dataplus, glossary, ouwiki, etc.
week_from	Integer	Week from which the resource is available.
week_to	Integer	Week until which the resource is available.

1.4 studentAssessment.csv

Contains scores students achieved on assessments, including coursework and exams.

Attribute	Type	Description
id_assessment	Integer	Assessment identifier.
id_student	Integer	Student identifier.
date_submitted	Integer	Day the assessment was submitted, relative to module start.
is_banked	Binary	1 if score was carried over from a previous attempt, 0 otherwise.
score	Continuous (0–100)	Score achieved by the student on the assessment.

1.5 assessments.csv

Metadata about each assessment within a module presentation.

Attribute	Type	Description
code_module	Categorical	Module identifier.
code_presentation	Categorical	Presentation identifier.

Attribute	Type	Description
id_assessment	Integer	Unique assessment identifier.
assessment_type	Categorical	Type: TMA (Tutor Marked Assignment), CMA (Computer Marked Assignment), or Exam.
date	Integer	Assessment due date relative to module start.
weight	Continuous	Weight of the assessment towards the final module score (%).

1.6 courses.csv

Provides general metadata about each module presentation.

Attribute	Type	Description
code_module	Categorical	Module identifier.
code_presentation	Categorical	Presentation identifier.
module_presentation_length	Integer	Length of the module in days.

1.7 studentRegistration.csv

Tracks when students registered and potentially unregistered from modules.

Attribute	Type	Description
code_module	Categorical	Module identifier.
code_presentation	Categorical	Presentation identifier.
id_student	Integer	Student identifier.
date_registration	Integer	Day of registration relative to module start (often negative).
date_unregistration	Integer	Day of unregistration; NULL if the student did not withdraw.

2. Operationalising the Hypothesis

The hypothesis is: 'Positive learning experiences contribute to sustained academic success.' To test this statistically, we must operationalise both constructs as measurable variables derived from the OULAD dataset.

2.1 Defining 'Positive Learning Experiences' (Independent Variable)

Positive learning experiences can be constructed as a composite measure from the following attributes:

Proxy Measure	Source File	Attributes Used	Justification
VLE Engagement Score	studentVle.csv + vle.csv	sum_click (aggregated per student), activity_type	High interaction frequency signals active learning and motivation — core indicators of positive experience.
Early Engagement	studentVle.csv	date (early weeks), sum_click	Students who engage before/early in a module show proactive behaviour, associated with positive outlook.
Assessment Submission Timeliness	studentAssessment.csv + assessments.csv	date_submitted vs date (due)	Submitting on time or early reflects positive engagement and reduced academic anxiety.
Assessment Performance (early)	studentAssessment.csv	score (TMA/CMA only, early assessments)	Good early scores reinforce motivation — a hallmark of positive learning experience.
Forum/Collaborative Activity	studentVle.csv + vle.csv	activity_type = 'forumng', sum_click	Forum participation reflects social learning — collaborative experiences are inherently positive.
Low num_of_prev_attempts	studentInfo.csv	num_of_prev_attempts	Students on first attempt have not yet experienced failure — inversely relates to adverse experience.

2.2 Defining 'Sustained Academic Success' (Dependent Variable)

Sustained academic success is not just a pass — it encompasses consistent performance across the module duration:

Proxy Measure	Source File	Attributes Used	Justification
Final Result	studentInfo.csv	final_result (Pass/Distinction)	Primary academic outcome measure. Distinction and Pass represent success; Fail and Withdrawn represent non-success.

Proxy Measure	Source File	Attributes Used	Justification
Assessment Score Trajectory	studentAssessment.csv	score over time (using date_submitted)	Sustained success = consistently high scores, not just a single good result. A positive slope indicates improvement.
Module Completion	studentRegistration.csv + studentInfo.csv	date_unregistration (NULL = completed), final_result != Withdrawn	Students who do not withdraw demonstrate sustained engagement — a prerequisite for success.
Credit Load Management	studentInfo.csv	studied_credits, final_result	Successfully managing high credit loads while achieving good results reflects sustained academic performance.

2.3 Engineered Variables (Python-Derived)

The following computed variables should be engineered in Python during data preparation:

Variable Name	Formula / Logic	Purpose
total_clicks	SUM(sum_click) per student per module	Overall VLE engagement score
early_clicks	SUM(sum_click) where date < 0.25 × module_length	Early engagement proxy
avg_score_TMA	MEAN(score) where assessment_type='TMA'	Coursework performance
submission_delay	MEAN(date_submitted - due_date) per student	Timeliness of submissions (negative = early)
forum_clicks	SUM(sum_click) where activity_type='forumng'	Social/collaborative engagement
positive_exp_score	Composite: standardised sum of above proxies	Single independent variable index
success_binary	1 if final_result in [Pass, Distinction], 0 otherwise	Binary dependent variable for logistic regression
score_trend	OLS slope of score vs assessment date per student	Trajectory of academic performance
completed	1 if date_unregistration is NULL AND final_result != Withdrawn	Module completion flag

3. Statistical Framework

3.1 Hypotheses

Formally, we define:

Null Hypothesis (H_0): Positive learning experiences have no statistically significant effect on sustained academic success. In other words, the mean academic outcomes of students with high vs. low positive experience scores are equal.

Alternative Hypothesis (H_1): Students with higher positive learning experience scores achieve significantly better sustained academic success outcomes.

Mathematical formulation:

```
H₀: μ_success_highPLE = μ_success_lowPLE  (or β_PLE = 0 in regression)
H₁: μ_success_highPLE > μ_success_lowPLE  (one-tailed, or β_PLE ≠ 0 two-tailed)
```

3.2 Significance Level & Error Types

Parameter	Value/Definition	Justification
Significance Level (α)	0.05 (5%)	Standard threshold in social/education research. Balances Type I and Type II errors.
Type I Error (α)	Rejecting H_0 when it is true — concluding engagement matters when it doesn't.	Risk of false positive. Controlled at 5%.
Type II Error (β)	Failing to reject H_0 when it is false — missing a real effect.	Risk of false negative. Managed via adequate sample size.
Statistical Power (1−β)	Target ≥ 0.80 (80%)	Probability of correctly detecting a real effect if it exists.
Confidence Interval	95% CI for all estimates	Range within which the true parameter lies with 95% probability.
p-value	Probability of observed data given H_0 is true	$p < 0.05$ leads to rejection of H_0.
Effect Size	Cohen's d (t-tests), Cramér's V (chi-square), R^2 (regression)	Quantifies the magnitude of the relationship beyond statistical significance.

4. Descriptive Analytics

Descriptive analytics summarises and characterises the dataset to understand distributions, central tendencies, and relationships before formal testing. This covers LO1 (probability distributions) and LO4 (exponential family distributions).

4.1 Univariate Analysis

Variable	Distribution Family	Analysis to Perform	Python Method
total_clicks	Poisson / Negative Binomial (count data)	Mean, median, std dev, skewness, histogram, Q-Q plot	scipy.stats.describe(), sns.histplot()
avg_score_TMA	Normal (continuous, bounded)	Mean, std dev, normality test (Shapiro-Wilk), histogram	scipy.stats.shapiro(), sns.histplot()
final_result	Categorical (4 classes)	Frequency table, bar chart, proportion by category	pd.value_counts(), sns.countplot()
num_of_prev_attempts	Poisson (discrete count)	Frequency distribution, mode, Poisson fit	scipy.stats.poisson.fit()
studied_credits	Approximately Normal	Mean, std dev, histogram	df['studied_credits'].describe()
submission_delay	Normal (can be negative)	Mean, std dev, box plot	sns.boxplot()

4.2 Bivariate & Multivariate Analysis

Variables	Analysis	Python Code Pattern
total_clicks vs final_result	Box plot of clicks by result category; ANOVA pre-test	sns.boxplot(x='final_result', y='total_clicks')
avg_score_TMA vs success_binary	Point-biserial correlation; scatter with regression line	scipy.stats.pointbiserialr(score, success)
positive_exp_score distribution by gender/region	Grouped descriptive stats; bar charts	df.groupby('gender')['positive_exp_score'].describe()
Forum engagement vs final_result	Crosstab; stacked bar chart	pd.crosstab(forum_engagement_band, final_result)
early_clicks vs score_trend	Pearson correlation; scatter plot	scipy.stats.pearsonr(early_clicks, score_trend)
imd_band vs success_binary	Grouped proportions; bar chart by deprivation band	df.groupby('imd_band')['success_binary'].mean()

4.3 Distribution Fitting (LO1 & LO4)

Fitting distributions to key variables establishes their probabilistic nature, which underpins all inferential testing:

- total_clicks → Fit Poisson distribution. Test: Chi-square goodness-of-fit. If overdispersed, use Negative Binomial.
- avg_score_TMA → Fit Normal distribution. Test: Shapiro-Wilk (n<2000) or Kolmogorov-Smirnov (larger samples).
- success_binary → Binomial distribution (n trials, p = proportion passing).
- Time between assessment submissions → Exponential/Gamma distribution (time-to-event).

```
from scipy import stats
# Poisson fit for click counts
lambda_est = df['total_clicks'].mean()
ks_stat, p_val = stats.kstest(df['total_clicks'], 'poisson', args=(lambda_est,))
# Normal fit for scores
mu, sigma = stats.norm.fit(df['avg_score_TMA'])
stat, p = stats.shapiro(df['avg_score_TMA'].dropna().sample(2000))
```

5. Inferential Analytics

Inferential analytics allows us to draw conclusions about the population from our sample. This directly addresses LO2 (hypothesis testing) and LO4 (exponential family distributions).

5.1 Test Selection Framework

Research Question	Test	Dependent Var	Independent Var	Assumption
Do high-engagement students pass more?	Independent samples t-test or Mann-Whitney U	avg_score_TMA	Engagement group (high/low)	Normality or large n
Is VLE engagement related to final result?	One-Way ANOVA (+ Tukey post-hoc)	total_clicks	final_result (4 groups)	Homogeneity of variance
Is there association between engagement level and result?	Chi-Square Test of Independence	Result category	Engagement band (low/med/high)	Expected freq ≥ 5
Does submission timeliness predict success?	Logistic Regression	success_binary	submission_delay	Independence, no multicollinearity
Are score means different before/after high engagement?	Paired t-test or Wilcoxon signed-rank	score at early vs late assessment	Within-student comparison	Paired data
Does deprivation (IMD) moderate the relationship?	Two-Way ANOVA or interaction term in regression	avg_score_TMA	PLE score + imd_band	Factorial design

5.2 Primary Test: Mann-Whitney U / Independent t-test

Compare academic scores between students with high vs. low Positive Learning Experience (PLE) scores:

```
from scipy import stats
high_PLE = df[df['positive_exp_score'] >
df['positive_exp_score'].median()]['avg_score_TMA']
low_PLE  = df[df['positive_exp_score'] <=
df['positive_exp_score'].median()]['avg_score_TMA']
# Check normality first
stat, p_norm = stats.shapiro(high_PLE.sample(min(2000, len(high_PLE))))
# If normal: t-test
t_stat, p_val = stats.ttest_ind(high_PLE, low_PLE, alternative='greater')
# If not normal: Mann-Whitney U
u_stat, p_val = stats.mannwhitneyu(high_PLE, low_PLE, alternative='greater')
```

5.3 ANOVA: VLE Engagement Across Result Groups

```
from scipy.stats import f_oneway
groups = [df[df['final_result']==r]['total_clicks'] for r in
['Distinction','Pass','Fail','Withdrawn']]
f_stat, p_val = f_oneway(*groups)
```

```
# Post-hoc Tukey HSD
from statsmodels.stats.multicomp import pairwise_tukeyhsd
tukey = pairwise_tukeyhsd(df['total_clicks'], df['final_result'])
```

5.4 Chi-Square Test: Engagement Band vs. Result

```
import pandas as pd; from scipy.stats import chi2_contingency
df['eng_band'] = pd.qcut(df['total_clicks'], q=3, labels=['Low','Mid','High'])
contingency = pd.crosstab(df['eng_band'], df['final_result'])
chi2, p, dof, expected = chi2_contingency(contingency)
# Cramér's V for effect size
import numpy as np
cramers_v = np.sqrt(chi2 / (contingency.sum().sum() * (min(contingency.shape)-1)))
```

5.5 Bayesian Analysis (LO6)

We apply Bayesian inference to update our beliefs about the probability that high-PLE students succeed, given observed data:

```
# Prior: Beta(α=2, β=2) — weakly informative, centered at 0.5
# Likelihood: Binomial (each student either succeeds or not)
# Posterior: Beta(α + successes, β + failures)
successes_high = df[df['positive_exp_score']>median]['success_binary'].sum()
n_high = df[df['positive_exp_score']>median].shape[0]
alpha_post = 2 + successes_high
beta_post  = 2 + (n_high - successes_high)
posterior_mean = alpha_post / (alpha_post + beta_post)
# 95% Credible Interval
from scipy.stats import beta as beta_dist
ci = beta_dist.ppf([0.025, 0.975], alpha_post, beta_post)
```

6. Predictive Analytics

Predictive analytics builds models that forecast outcomes from input features. This covers LO3 (regression), LO5 (time series), and LO6 (Bayesian methods).

6.1 Logistic Regression (LO3) — Primary Predictive Model

Logistic regression models the probability of sustained academic success (binary outcome) as a function of positive learning experience predictors.

Variable	Role	Type
success_binary	Dependent (Y)	Binary: 1=Pass/Distinction, 0=Fail/Withdrawn
total_clicks	Independent (X_1)	Continuous — overall VLE engagement
avg_score_TMA	Independent (X_2)	Continuous — coursework performance
early_clicks	Independent (X_3)	Continuous — early engagement
forum_clicks	Independent (X_4)	Continuous — collaborative engagement
submission_delay	Independent (X_5)	Continuous — timeliness
num_of_prev_attempts	Control variable	Discrete — prior experience
studied_credits	Control variable	Continuous — workload
imd_band	Control variable	Ordinal — socioeconomic status

```
import statsmodels.api as sm
features = ['total_clicks','avg_score_TMA','early_clicks','forum_clicks',
            'submission_delay','num_of_prev_attempts','studied_credits']
X = sm.add_constant(df[features])
y = df['success_binary']
model = sm.Logit(y, X).fit()
print(model.summary())
# Odds ratios
odds_ratios = np.exp(model.params)
# Marginal effects
marginal_effects = model.get_margeff()
```

6.2 Multiple Linear Regression (LO3) — Score Prediction

Linear regression predicts continuous assessment score from engagement variables:

```
X =
sm.add_constant(df[['total_clicks','early_clicks','forum_clicks','submission_delay']])
y = df['avg_score_TMA']
ols_model = sm.OLS(y, X).fit()
print(ols_model.summary())  # R², F-stat, coefficients, p-values
# Assumptions checks
from statsmodels.stats.diagnostic import het_breuschpagan
bp_test = het_breuschpagan(ols_model.resid, X)  # Heteroscedasticity
from statsmodels.stats.stattools import durbin_watson
dw = durbin_watson(ols_model.resid)  # Autocorrelation
```

6.3 Time Series Analysis (LO5)

Using assessment scores ordered by submission date, we can model academic performance as a time series per student. This captures the 'sustained' dimension of the hypothesis.

```
import statsmodels.tsa.holtwinters as ets
# Aggregate average score per assessment date across all students
ts_data = df_merged.groupby('assessment_date')['score'].mean().sort_index()
# Holt-Winters Exponential Smoothing for trend
hw_model = ets.ExponentialSmoothing(ts_data, trend='add').fit()
forecast = hw_model.forecast(steps=5)
# ARIMA for score trend modelling
from statsmodels.tsa.arima.model import ARIMA
arima = ARIMA(ts_data, order=(1,1,1)).fit()
print(arima.summary())
```

6.4 Model Evaluation

Model	Evaluation Metrics	Acceptable Threshold
Logistic Regression	Accuracy, AUC-ROC, Confusion Matrix, Classification Report	AUC > 0.70 indicates good discriminatory power
Linear Regression (OLS)	R^2, Adjusted R^2, RMSE, F-statistic	$R^2 > 0.30$ for educational data; $p(F) < 0.05$
Time Series (Holt-Winters)	MAPE, RMSE on test holdout	MAPE < 15% for acceptable forecast accuracy
All models	k-fold Cross Validation (k=5 or 10)	Stable CV score ± small std indicates generalisation

7. Justification for Research Topic & Dataset Choice

7.1 Why This Topic?

Business Perspective

Educational institutions operate as service organisations with financial imperatives. Student dropout is costly — universities lose tuition fees, government funding tied to completion rates, and institutional reputation. Understanding what constitutes positive learning experiences and their link to success allows institutions to design better interventions, reducing dropout rates and improving return on educational investment. EdTech companies (learning management system providers, adaptive learning platforms) can use such insights to commercialise evidence-based engagement tools worth billions in the global EdTech market.

Education Perspective

The hypothesis directly addresses a core pedagogical debate: does how students engage with learning material determine outcomes beyond innate ability? This is central to educational psychology theories such as Vygotsky's Zone of Proximal Development and Bandura's Self-Efficacy Theory. Empirically validating this with real data moves the debate from theory to evidence-based policy. It supports curriculum designers in optimising resource placement, assessment scheduling, and forum facilitation.

Societal Perspective

The OULAD dataset includes the Index of Multiple Deprivation (IMD) band, allowing us to examine whether positive learning experiences moderate the disadvantage faced by students from lower socioeconomic backgrounds. If engagement mitigates socioeconomic disparity in outcomes, it has profound implications for educational equity policy, government funding decisions, and social mobility initiatives. At a time when lifelong learning and adult education are critical societal priorities, understanding what sustains success for diverse learners is of enormous public value.

Personal / Team Perspective

As students currently engaged in an educational institution ourselves, this topic is personally resonant. We have direct experiential understanding of what constitutes a positive learning experience — effective feedback, accessible resources, collaborative discussion — and what sustained engagement feels like. This personal connection motivates rigorous analysis and helps us interpret results with contextual authenticity. The topic also develops transferable data science and statistical skills directly applicable to our future careers in data analytics, business intelligence, and educational research.

7.2 Why OULAD Rather Than Other Datasets?

Criterion	OULAD	Alternative Datasets
Size & Statistical Power	32,593 students, 7 tables, millions of VLE interaction records — large n ensures high statistical power.	Many education datasets have <1000 observations, insufficient for multiple regression and time series.
Richness of Variables	Covers demographics, engagement behaviour (clickstream), assessments, and outcomes — all required to construct PLE and success measures.	Kaggle student datasets often have only grades and demographics — no engagement data.
Temporal Granularity	Daily VLE interaction logs enable time series analysis (LO5) and trajectory modelling.	Most datasets provide only final scores — no temporal dimension for time series.

Criterion	OULAD	Alternative Datasets
Public Availability & Reproducibility	Freely available from analyse.kmi.open.ac.uk — fully documented, peer-reviewed, widely cited in 200+ academic papers.	Many alternative datasets lack proper documentation or have restricted access.
Alignment with Learning Outcomes	Supports Normal, Binomial, Poisson distributions (LO4), regression (LO3), time series (LO5), and Bayesian analysis (LO6).	Most datasets support only 1–2 of our required analyses.
Ethical Compliance	Fully anonymised by Open University, GDPR-compliant, no re-identification risk.	Primary collection or scraping would require IRB/ethics approval, informed consent, data storage protocols.

7.3 Why Not Primary Data Collection?

While primary data collection would offer maximum control over variable operationalisation, we deliberately chose OULAD as a secondary dataset for the following reasons:

1. Time constraints: Designing a survey, obtaining ethics approval, recruiting participants, collecting data, and cleaning it would consume more time than the assignment window allows.
2. Sample size: A student-administered survey would likely yield fewer than 200 responses — far too small for logistic regression, ANOVA with multiple groups, and time series analysis simultaneously.
3. Temporal data impossibility: Capturing daily learning engagement data over a full academic module would require months of passive data logging — not feasible in a coursework timeframe.
4. Ethical complexity: Collecting data about student performance and engagement from fellow students involves informed consent, anonymisation, institutional approval, and potential conflicts of interest.
5. Reproducibility and validity: OULAD has been validated, used in peer-reviewed research, and has known data quality. Our primary data would be unvalidated and potentially biased by self-selection and social desirability.
6. Alignment with real analytics practice: In professional data analytics, analysts more often work with existing institutional datasets than conduct primary surveys. Using OULAD simulates realistic industry conditions.

8. Team Work Plan & GitHub Branch Strategy

The team consists of 4 members. Work is divided by analytical phase to align with the assignment components and ensure each member covers distinct learning outcomes. Tasks follow the order: data → descriptive → inferential → predictive → reporting.

8.1 Member Roles Overview

Member	Primary Role	LOs Covered	GitHub Branch
Member 1 (Data Lead)	Data Collection, Cleaning & EDA Setup	LO1, LO4	feature/data-pipeline
Member 2 (Inferential Analyst)	Hypothesis Testing & Statistical Inference	LO2, LO4	feature/inferential-analysis
Member 3 (Predictive Modeller)	Regression, Time Series & Forecasting	LO3, LO5	feature/predictive-models
Member 4 (Bayesian & Reporting Lead)	Bayesian Analysis, Interpretation & Final Report	LO2, LO6	feature/bayesian-reporting

8.2 Detailed Task Breakdown

Member 1 — Data Pipeline (Branch: feature/data-pipeline)

Task	Sub-task	Mini-tasks	Output
T1.1: Data Acquisition	Download all 7 OULAD CSV files	Verify file integrity; check row counts; confirm column names match documentation	Raw data folder /data/raw/
T1.2: Data Merging	Join studentInfo + studentVle + assessments + studentAssessment	Merge on id_student + code_module + code_presentation; handle one-to-many relationships; validate join cardinality	df_merged.csv
T1.3: Data Cleaning	Handle missing values and outliers	Identify NULLs in date_unregistration, score, imd_band; impute or flag; detect outliers in total_clicks using IQR; document cleaning decisions	df_clean.csv
T1.4: Feature Engineering	Compute all derived variables	Calculate total_clicks, early_clicks, forum_clicks, avg_score_TMA, submission_delay, positive_exp_score (z-score composite), success_binary, score_trend (OLS slope per student)	df_features.csv
T1.5: Distribution Analysis	Fit distributions to key variables	Shapiro-Wilk for scores; KS test for clicks vs Poisson; Gamma fit for time-to-submission; generate Q-Q plots and histograms	EDA notebook: 01_eda.ipynb

Task	Sub-task	Mini-tasks	Output
T1.6: Descriptive Statistics	Summary statistics and visualisations	Compute mean/median/std/skew for all continuous variables; frequency tables for categorical; bar charts, box plots, correlation heatmap	Figures folder: /figures/descriptive/

Member 2 — Inferential Analysis (Branch: feature/inferential-analysis)

Task	Sub-task	Mini-tasks	Output
T2.1: Hypothesis Formulation	Formally state H_0 and H_1	Write mathematical expressions; define α=0.05; specify one-tailed vs two-tailed; document in notebook	hypothesis_statement.md
T2.2: Assumption Testing	Verify test assumptions before proceeding	Levene's test for equal variances; Shapiro-Wilk for normality; check expected frequencies for chi-square; VIF for multicollinearity	02_assumptions.ipynb
T2.3: t-test / Mann-Whitney	Compare success metrics by PLE group	Median-split positive_exp_score into high/low; run independent t-test; if non-normal run Mann-Whitney U; compute Cohen's d; interpret results	02_ttest.ipynb
T2.4: One-Way ANOVA	Test VLE engagement differences across result groups	Group total_clicks by final_result (4 levels); run ANOVA; Tukey HSD post-hoc; box plots per group; report F-stat, p-value, η^2	02_anova.ipynb
T2.5: Chi-Square Test	Test association between engagement band and result	Bin total_clicks into tertiles; build contingency table; run chi2_contingency; compute Cramér's V; visualise with mosaic plot	02_chisquare.ipynb
T2.6: Confidence Intervals	Compute CIs for key proportions	Bootstrap or analytical 95% CI for proportion passing in each engagement band; interpret overlap; report in summary table	02_confidence_intervals.ipynb

Member 3 — Predictive Models (Branch: feature/predictive-models)

Task	Sub-task	Mini-tasks	Output
T3.1: Data Split	Prepare training and test sets	80/20 train-test split stratified on success_binary and code_module; set random seed=42; check class balance	03_data_split.ipynb
T3.2: OLS Regression	Predict avg_score_TMA from PLE variables	Build model with statsmodels OLS; check R^2, adjusted R^2, F-stat, coefficient p-values; residual plots; Breusch-Pagan test; Durbin-Watson	03_ols_regression.ipynb

Task	Sub-task	Mini-tasks	Output
T3.3: Logistic Regression	Predict success_binary	Fit logit model; check odds ratios; AUC-ROC curve; confusion matrix; classification report; 5-fold cross-validation	03_logistic_regression.ipynb
T3.4: Model Diagnostics	Validate regression assumptions	Check multicollinearity (VIF < 5); normality of residuals (Q-Q plot); homoscedasticity; influential points (Cook's distance)	03_model_diagnostics.ipynb
T3.5: Time Series Setup	Aggregate score data as time series	Group assessment scores by date across all students; handle irregular intervals; interpolate if needed; plot ACF and PACF	03_timeseries_prep.ipynb
T3.6: Time Series Modelling	Fit ARIMA and Holt-Winters models	Determine ARIMA order via AIC/BIC; fit Holt-Winters with additive trend; forecast 5 periods ahead; evaluate with MAPE and RMSE on holdout	03_timeseries_model.ipynb

Member 4 — Bayesian Analysis & Reporting (Branch: feature/bayesian-reporting)

Task	Sub-task	Mini-tasks	Output
T4.1: Bayesian Prior Specification	Define and justify priors	Select Beta(2,2) weakly informative prior; document reasoning; plot prior distribution; justify choice vs flat prior	04_bayesian_prior.ipynb
T4.2: Bayesian Inference	Compute posterior distributions	Update prior with observed success/failure counts for high-PLE group; compute Beta posterior; derive posterior mean and 95% credible interval; compare to frequentist p-value	04_bayesian_inference.ipynb
T4.3: Bayesian Model Comparison	Compare models with different priors	Test sensitivity to prior: flat Beta(1,1) vs informative Beta(5,2); report how posterior changes; Bayes Factor calculation	04_bayes_factor.ipynb
T4.4: Result Synthesis	Integrate findings from all analyses	Compile descriptive, inferential, predictive, and Bayesian results into coherent narrative; build evidence table; identify converging findings	synthesis_table.md
T4.5: Visualisation Polish	Finalise all figures	Standardise plot styles (seaborn theme, consistent colours); label axes; add titles and statistical annotations; export high-res PNGs	/figures/final/
T4.6: Report Writing	Write final report	Write introduction, methodology, results, discussion, conclusion sections; cite all tests with	final_report.pdf

Task	Sub-task	Mini-tasks	Output
		statistics; format tables; proofread and submit	

8.3 Execution Order & Dependencies

Phase	Tasks	Who	Depends On	Estimated Time
Phase 1: Data	T1.1 → T1.4	Member 1	Nothing (starts first)	Week 1
Phase 2: EDA	T1.5 → T1.6	Member 1	Phase 1 complete	Week 1–2
Phase 3: Inference	T2.1 → T2.6	Member 2	df_features.csv from Phase 1	Week 2
Phase 4: Prediction	T3.1 → T3.6	Member 3	df_features.csv from Phase 1	Week 2–3
Phase 5: Bayesian	T4.1 → T4.3	Member 4	df_features.csv from Phase 1	Week 2–3
Phase 6: Integration	T4.4 → T4.6	Member 4 (all review)	Phases 3, 4, 5 complete	Week 3–4

8.4 GitHub Repository Structure

```
main                          ← Protected: final merged code only
develop                       ← Integration branch: merge feature branches here first
feature/data-pipeline         ← Member 1: all data tasks
feature/inferential-analysis  ← Member 2: all hypothesis testing
feature/predictive-models     ← Member 3: all regression & time series
feature/bayesian-reporting    ← Member 4: Bayesian + report writing

Workflow: feature/* → PR to develop → review → merge → final PR to main
```

8.5 Shared Responsibilities (All Members)

- Attend all team meetings and update progress on shared project board.
- Review and comment on at least one pull request from another member each week.
- Follow coding standards: PEP8, docstrings, markdown explanations in notebooks.
- Contribute to viva preparation: each member presents their own section.

9. Python Implementation Guide

9.1 Required Libraries

```
pip install pandas numpy scipy statsmodels scikit-learn matplotlib seaborn pymc
```

9.2 Data Loading Template

```
import pandas as pd
student_info    = pd.read_csv('studentInfo.csv')
student_vle     = pd.read_csv('studentVle.csv')
vle             = pd.read_csv('vle.csv')
student_assess  = pd.read_csv('studentAssessment.csv')
assessments     = pd.read_csv('assessments.csv')
courses         = pd.read_csv('courses.csv')
registration    = pd.read_csv('studentRegistration.csv')
```

9.3 Feature Engineering Template

```
# Total clicks per student per module
click_agg = student_vle.groupby(['id_student','code_module','code_presentation'])
           ['sum_click'].sum().reset_index().rename(columns={'sum_click':'total_clicks'})

# Early clicks: first quarter of module
merged_vle = student_vle.merge(courses, on=['code_module','code_presentation'])
early_mask = merged_vle['date'] < (merged_vle['module_presentation_length'] * 0.25)
early_clicks =
merged_vle[early_mask].groupby(['id_student','code_module'])['sum_click'].sum()

# Forum clicks
forum_vle = student_vle.merge(vle[['id_site','activity_type']], on='id_site')
forum_clicks = forum_vle[forum_vle['activity_type']=='forumng']
             .groupby(['id_student','code_module'])['sum_click'].sum()

# Binary success variable
student_info['success_binary'] =
student_info['final_result'].isin(['Pass','Distinction']).astype(int)

# Composite PLE score (z-score standardise and sum)
from scipy.stats import zscore
ple_cols = ['total_clicks','early_clicks','forum_clicks','avg_score_TMA']
df[ple_cols] = df[ple_cols].apply(zscore)
df['positive_exp_score'] = df[ple_cols].mean(axis=1)
```